Name: Ohide Andrew Odohu
Date: 16/10/2026
Course/Class Title: Software engineering

1. Task 1 - Flutter & Dart Setup

```
PS C:\Users\USER> where.exe flutter
C:\Users\USER\flutter\bin\flutter
C:\Users\USER\flutter\bin\flutter.bat
PS C:\Users\USER> flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.47.7, on Microsoft Windows [Version 10.0.26200.9711], locale
en-US)
[√] Windows Version (Windows 11 or higher, 25H2, 2009)
[X] Android toolchain - develop for Android devices
    X Unable to locate Android SDK.
      Install Android Studio from: https://developer.android.com/studio/index.html
      On first launch it will assist you in installing the Android SDK components.
      (or visit https://flutter.dev/to/windows-android-setup for detailed instructions).
      If the Android SDK has been installed to a custom location, please use
      `flutter config --android-sdk` to update to that location.

[√] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default
components
[√] Connected device (3 available)
[√] Network resources

! Doctor found issues in 2 categories.
PS C:\Users\USER>
```

```
PS C:\Users\USER\Desktop\PLP ACADEMY\my_first_app\my_first_app> flutter create
my_first_app
Recreating project my_first_app...
Resolving dependencies in `my_first_app`... (1.1s)
Downloading packages...
Got dependencies in `my_first_app`.
Wrote 3 files.

All done!
You can find general documentation for Flutter at:
https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider:
```

4. Task 4 - VS Code Workspace Screenshot

https://www.youtube.com/c/flutterdev

In order to run your application, type:

```
$ cd my_first_app
$ flutter run
```

Your application code is in my_first_app\lib\main.dart.

```
PS C:\Users\USER\Desktop\PLP ACADEMY\my_first_app\my_first_app> cd my_first_app
PS C:\Users\USER\Desktop\PLP ACADEMY\my_first_app\my_first_app\my_first_app>
flutter devices
Found 3 connected devices:
  Windows (desktop) • windows • windows-x64    • Microsoft Windows
  [Version 10.0.26200.9711]
  Chrome (web)      • chrome  • web-javascript • Google Chrome
  155.0.8059.40
  Edge (web)        • edge    • web-javascript • Microsoft Edge
  155.0.4283.45

Run "flutter emulators" to list and start any available device
emulators.

If you expected another device to be detected, please run "flutter
doctor" to diagnose potential issues. You may also try increasing the
time to wait for connected devices with the "--device-timeout" flag.
Visit https://flutter.dev/setup/ for troubleshooting tips.
PS C:\Users\USER\Desktop\PLP ACADEMY\my_first_app\my_first_app\my_first_app>
```

Hot Reload

Hot Reload allows a developer to quickly apply changes made to the Dart source code to a running Flutter application without restarting the entire application. The current application state is generally preserved. It is useful when making small changes to the user interface, styles, layouts, or other code where the existing application state should be maintained.

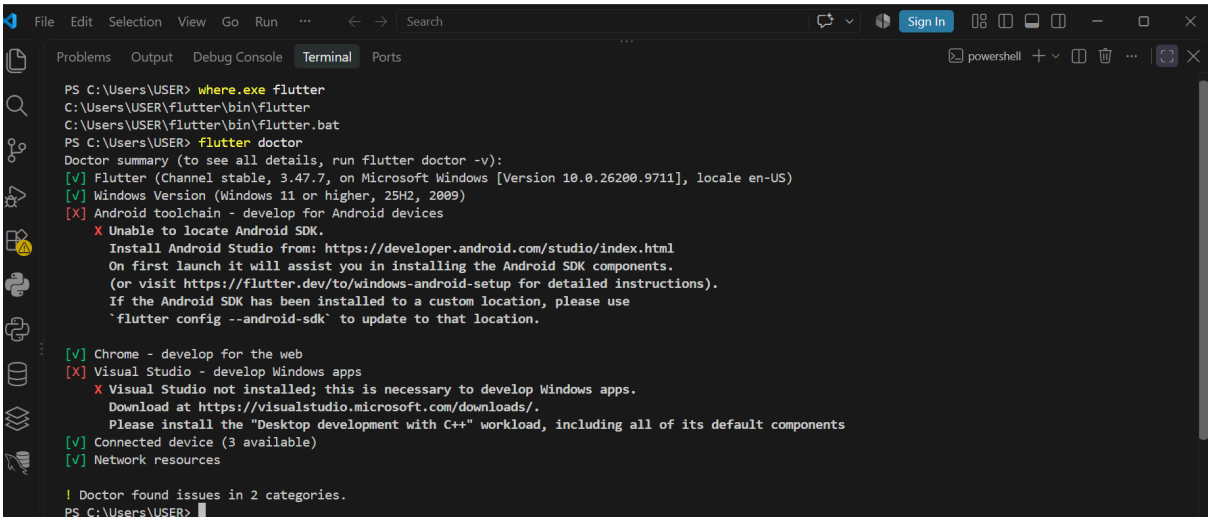
Hot Restart

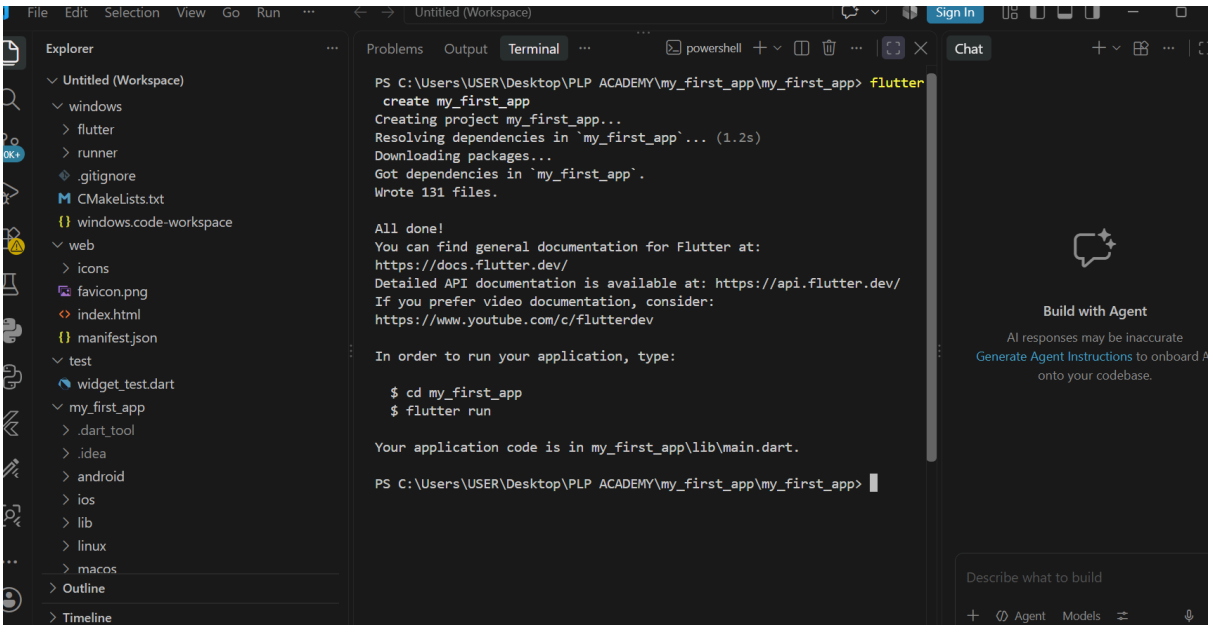
Hot Restart completely restarts the Flutter application and runs the Dart code again from the beginning. Unlike Hot Reload, the current application state is lost. Hot Restart is useful when changes affect application initialization, global state, or when Hot Reload does not correctly reflect the changes.

4. Task 4 - VS Code Workspace Screenshot

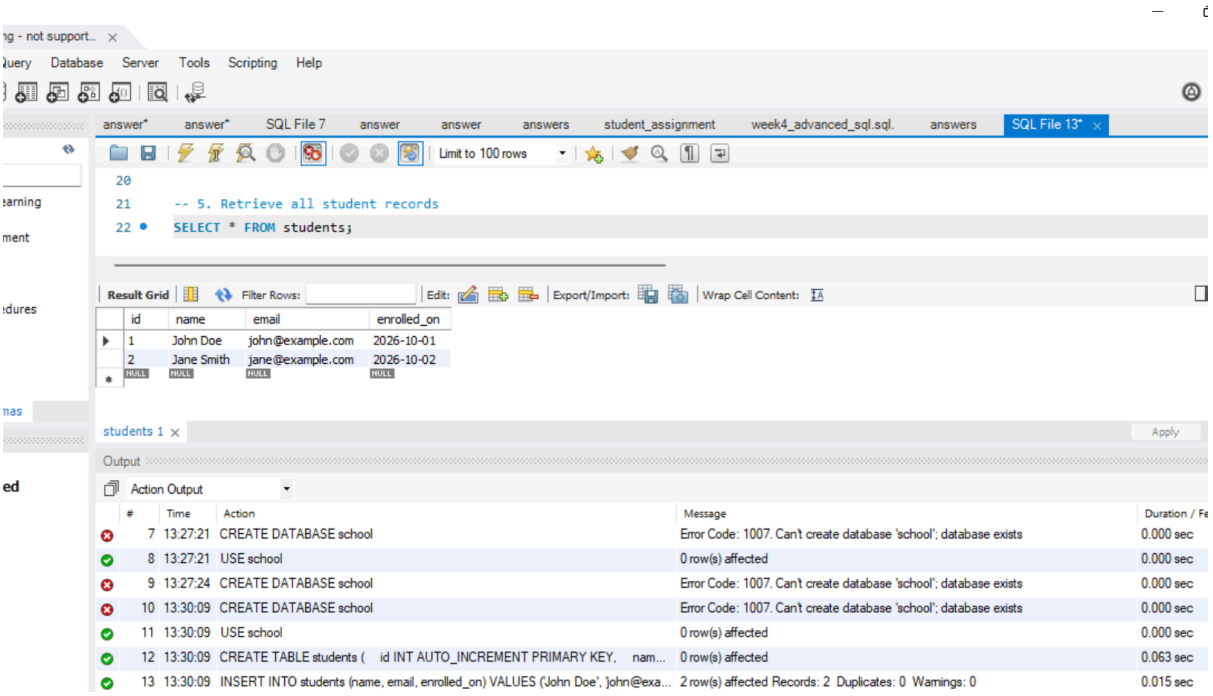
Simple comparison

Hot Reload	Hot Restart
Faster	Slower than Hot Reload
Preserves application state	Resets application state
Updates changed code	Restarts the application
Good for UI/code changes	Good when initialization/state must be reset





4. Task 4 - VS Code Workspace Screenshot

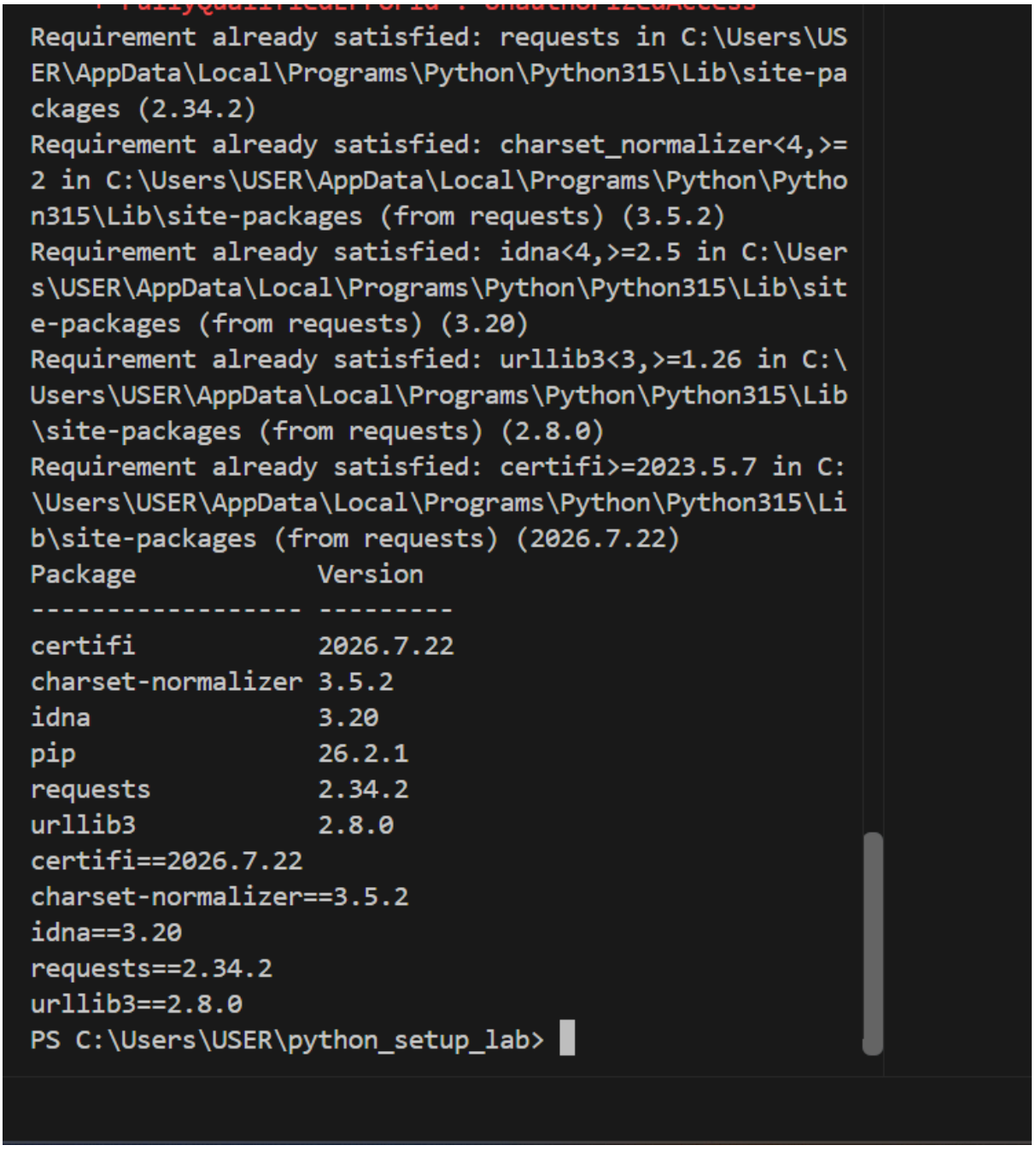


Security Reflection

Using the MySQL root account for an application's backend connection is poor security practice because root has extensive administrative privileges. If an attacker gains access to the root credentials, they could damage or delete databases, access sensitive information, or create unauthorized accounts.

To improve security, a dedicated application user should be created with only the permissions required to perform its tasks. This follows the principle of least privilege and reduces the potential damage if the account is compromised.

4. Task 4 - VS Code Workspace Screenshot



4. Task 4 - VS Code Workspace Screenshot

